
FOXA2 S169P Variant: Structural and Functional Implications in iPSC Differentiation

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Abstract

Induced pluripotent stem cells (iPSCs) are widely used in the field of regenerative medicine because of their ability to differentiate into different cell types. The iPSC differentiation is tightly related to transcription factors, which play important roles in cell fate determination. FOXA2 is a pioneer transcription factor that is involved in endoderm development and hepatic gene regulation. However, the structural and functional impact of the S169P variant remains unclear.

In this study, we performed a multi-level analysis on protein structure and interactions to evaluate the potential effects of the S169P mutation. We used AlphaFold models to predict the structure of S169P and all other alternative amino acids substitutions at residue 169. For DNA-binding test, we obtained the FOXA2-DNA binding model from Protein Data Bank (PDB), and investigated the location of residue 169. Protein-protein interaction networks were performed using STRING, analyzing the function of the interacting proteins and its potential protein interaction activity.

Our results indicate that S169P shows a relatively small effect on the protein structure, with the smallest root mean square deviation (RMSD) among all 19 mutants. Furthermore, residue S169 is surface-exposed and located outside the DNA-binding region, indicating a limited effect on DNA-binding affinity but a potential involvement in protein-protein interaction. These findings provided structural insight into the FOXA2 S169P variant and a foundation for future investigation in the iPSC context.

Keywords: iPSC, FOXA2, transcription factor, S169P variant, AlphaFold, Protein-protein interaction

1 Introduction

1.1 Research Background

Induced pluripotent stem cells (iPSCs) were first introduced in 2006 by two Japanese scientists: Shinya Yamanaka and Kazutoshi Takahashi at Kyoto University(1).

iPSCs are stem cells that can be generated directly from somatic cells. This characteristic enables them to differentiate into multiple cell types, including organ-specific cells, and to generate organoid models such as liver organoids(3). The mechanism involved in this process is primarily based on the regulation of transcription factors, and it has been widely applied to medical use in the field of regenerative medicine(1; 2)

Transcription factors (TFs) play an important role in iPSC regulation, as they determine the gene expression and affect downstream cellular functions. Previous studies have revealed that TFs are essential in driving cell fate specification. Therefore, it is critical to understand the mechanism

involved in this process, including its interactive proteins, DNA binding affinity, and structural mutation influence(4).

FOXA2 (Forkhead Box A2), a transcription factor belonging to the Forkhead box A family, functions as an essential regulator in the hepatic gene regulatory network, and acts as a key transcription factor for endoderm differentiation. It activates the developmental genes and induces the liver-specific gene expression. Thus, the deletion or loss of function of FOXA2 may lead to severe regulatory disruptions, and significantly influence downstream gene expression, contributing to disease development(5). In the context of iPSCs, the mutation of FOXA2 may lead to the inefficiency of endodermal cells production, disrupting the downstream cell specification.

1.2 Research Gap and Research Questions

Although FOXA2 has been widely recognized as key regulator, the structural and functional consequences of specific points mutations remain uncertain.

According to previous research, the FOXA2 S169P mutation, in which serine at position 169 is substituted by proline, is associated with severe diseases related to liver-specific gene expression. The research results demonstrated that patients with S169P mutation in FOXA2 commonly suffer from congenital hypopituitarism (CH) and many other developmental issues affecting blood glucose homeostasis and hormonal regulation(6).

However, the downstream functional consequences of S169P mutation in FOXA2 function remains poorly understood. It is unclear whether the mutation influences the protein structure, DNA binding affinity, or the interactions with other proteins.

To address this issue, this study aims to systematically investigate the S169P mutation impact, and to provide a mechanistic insight into its structural and functional implications in the iPSCs differentiation.

2 Results

2.1 Structure Modeling of FOXA2

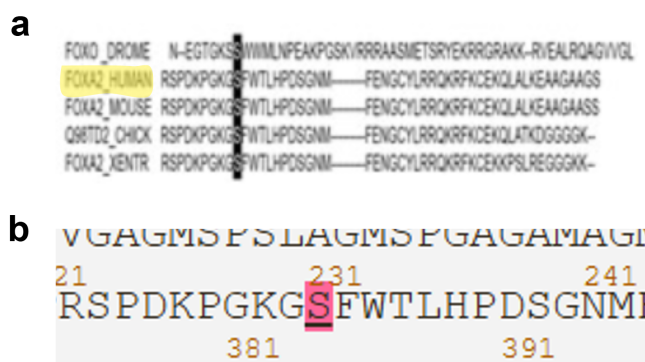


Figure 1: Sequence comparison

We first modeled the protein structure of human FOXA2 using AlphaFold Model V6(7). Due to recent discovery updates, the amino acid sequence of FOXA2 has been extended, and the previous residue annotated as position 169 is now numbered as position 231.

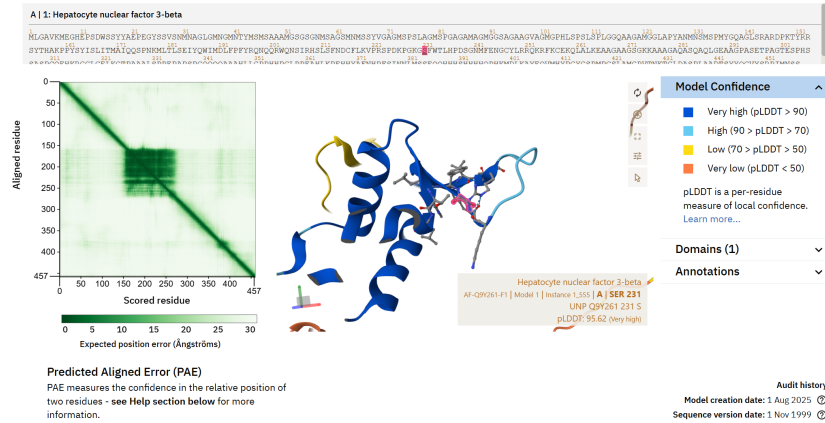


Figure 2: FOXA2 3D structure prediction with AlphaFold

The predicted aligned error (PAE) map was examined to evaluate the confidence of the FOXA2 structure. Overall, low APE values were observed within the structured regions, indicating high confidence level. Residue 231 is located within a region showing low PAE, suggesting that this position is relatively constrained.

The three-dimensional structure of FOXA2 has an average pLDDT of 56.69, indicating relatively overall low confidence. However, residue 231 locates in dark blue region (high pLDDT), suggesting high level of local confidence.

The structural prediction shows that the residue 231 is located within an α -helix, suggesting that this mutation may affect the secondary structure of the protein, and thereby disrupt downstream function.

Additionally, proline is known as the " α -helix breaker" due to its unique side chain, which cyclizes back to the backbone amide and makes it difficult to maintain helical structure in the protein center. This property makes it likely to destabilize the local helix structure, potentially leading to further protein folding and functional interaction problems(8).

2.2 Mutation structural comparison

To test our structural hypothesis, we aligned the S231P mutant structure with wild type structure using PyMOL.

We performed two structural alignments to evaluate local structure differences, one using a ± 100 amino acid (aa) window surrounding the mutation site and another using a ± 50 aa window.

We performed tests using different amino acid sequence length since AlphaFold predictions are optimized globally and may ignore localize perturbations caused by single residue mutations.

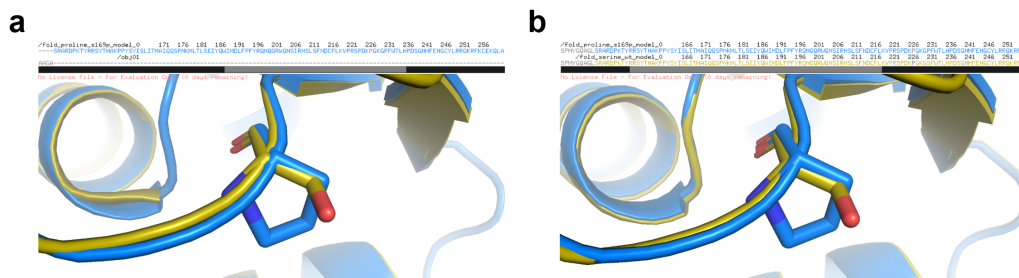


Figure 3: 100aa (S132-L330) and 50aa (L182-A280) alignment

83 The root mean square deviation (RMSD) from both structural alignments were low, with RMSD=0.321
84 for the ± 100 aa window surrounding mutation site, and RMSD=0.097 for the ± 50 aa window.

85 Interestingly, these results contradicts the usual phenomenon, it suggests that this mutation does not
86 significantly change the protein structure, and the functional consequences may arise from other
87 mechanisms.

88 In order to determine whether the observed effect is specific to proline substitution or reflects general
89 phenomenon, we modeled the remaining 18 single-amino acid substitutions using AlphaFold, and
90 aligned each mutant with the wild-type protein structure.

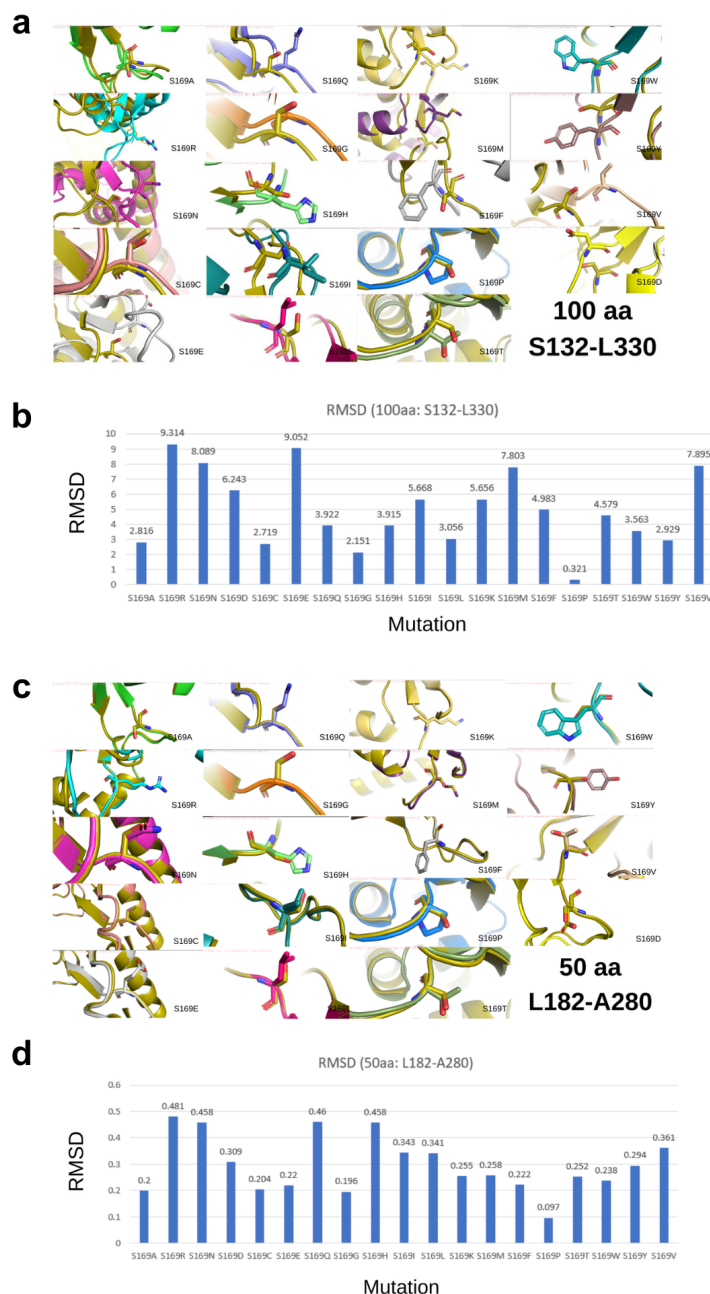


Figure 4: Mutant and wild-type structure comparison and RMSD of alignment

91 Structural alignment results provide evidence that the S231P exhibits the lowest RMSD among the
92 substitutions tested, suggesting that S231P is a relatively special mutant, and mechanism involved
93 may differ from that of the other mutants.

94 This indicates that S231P induces only minimal global structural change, and may represent a specific
95 effect.

96 **2.3 DNA binding mapping**

97 Based on the structural analysis above, we hypothesized that the functional consequences of the
98 FOXA2 S231P mutation may be caused by altered DNA-binding affinity.

99 Therefore, we obtained the crystal structure of DNA-binding domain of FOXA2 from the Protein
100 Data Bank (PDB)(9).

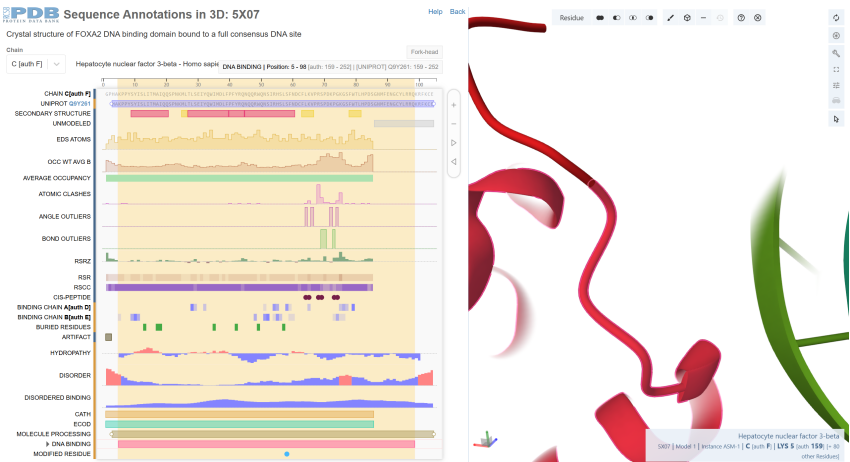


Figure 5: DNA binding sequence

101 The DNA-binding domain of FOXA2 is from residue 5-98, indicating that residue 231 isn't located
102 in the DNA-binding region.

103 Thus, the S231 mutation lies outside this region, and it is unlikely to directly affect the DNA binding.

104 Furthermore, we used PyMOL to examine the surface accessibility of residue S231. The AlphaFold-
105 predicted structure showed that this residue is surface-exposed, indicating a higher probability of
106 protein-protein interaction(10).

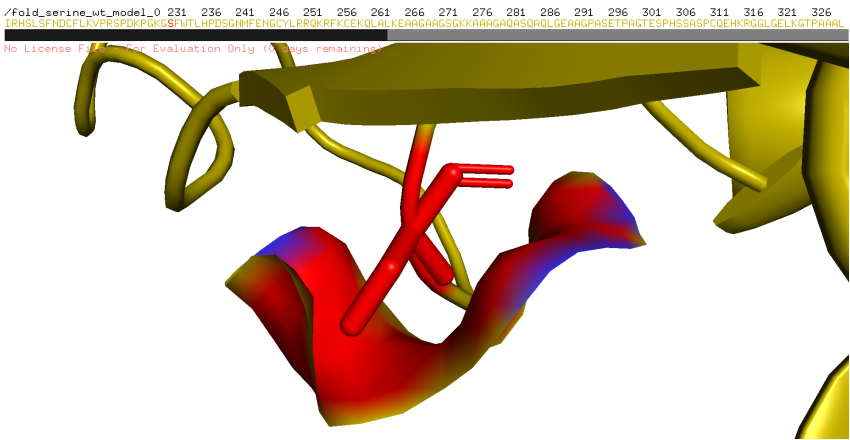


Figure 6: Residue S231 surface accessibility

2.4 Protein-protein interaction network

Since S231 mutation neither disturbs the overall protein folding and structure, or affects the DNA-binding affinity, we hypothesized that the functional consequences may arise from disruption of protein-protein interactions(PPIs)(11).

To explore potential mechanisms of the S231P functional impact, we used STRING to obtain the PPI network of FOXA2.

FOXA2 PPI network consists of 11 nodes (11 proteins), 31 edges (31 interactions), indicating a relatively dense interaction network.

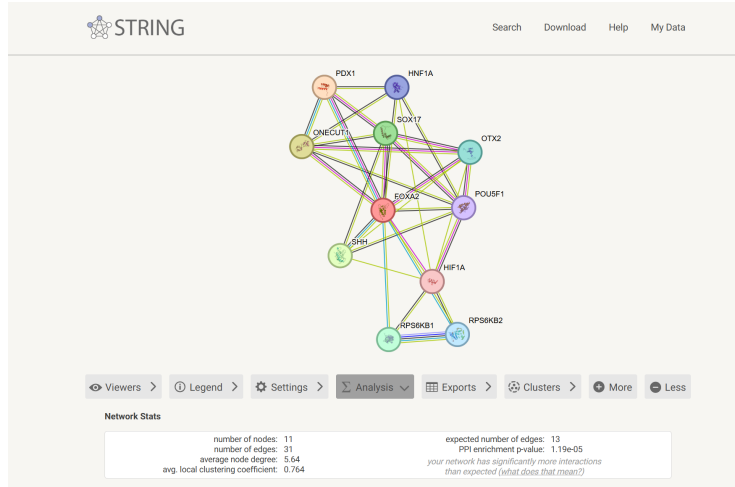


Figure 7: Interactive protein network

We summarized the functions of the interacting proteins, and found that a large proportion of them are involved in the regulation of genes related to pancreas, liver, and early development.

Table 1: Interactive proteins and function

Protein	Function
PDX1	Key transcription factor for pancreatic development and β -cell insulin regulation(12).
HNF1A	Regulates gene expression in liver, pancreas, kidney, controlling glucose metabolism(13).
ONECUT1	Drives liver and pancreatic endocrine cell differentiation(14).
SOX17	Controls endoderm formation and early organ lineage specification(15).
OTX2	Directs early embryonic development and regulates brain formation(16).
POU5F1	Maintains pluripotency and self-renewal in embryonic stem cells(17).
HIF1A	Mediates cellular response to hypoxia by activating survival and metabolic genes(18).
RPS6KB1	Promotes protein synthesis and cell growth through mTOR signaling(19).
RPS6KB2	Enhances protein synthesis and cell proliferation via protein S6 phosphorylation(20).

Among these proteins, those which are functionally associated with organ development (such as PDX1, HNF1A, ONECUT1, SOX17, OTX2), may have significant effects on downstream gene regulation, and influence developmental specification.

3 Discussion

In this study, we investigated the structural and functional implications of the FOXA2 S169P variant. Structural modeling suggested that the mutation does not lead to large-scale structural and protein

123 conformation changes. However, predicted protein structure indicated that residue S169 is surface-
124 exposed, giving rise to the possibility that S169 mutation may affect protein-protein interactions.
125 Taken together, these findings suggest that the S169P variant may influence FOXA2 regulatory
126 activity through altering the protein interaction rather than global structural.

127 **3.1 Structural influence analysis**

128 Results of structural analysis revealed that the S231 residue is located within the α -helix, making it
129 possible to disrupt the helical structure and affect further folding and protein stability. However, the
130 alignment of the mutant structure showed relatively low RMSD scores, indicating that the mutation
131 only altered the protein shape slightly in overall conformation.

132 Surprisingly, the S231P mutant exhibited an RMSD of 0.321 when retaining ± 100 aa (S132-L330)
133 surrounding the mutation site, and 0.097 when retaining ± 50 aa (L182-A280, yet it still leads to
134 significant functional impact.

135 These results indicates that the function consequences of S231P may be influenced by other mecha-
136 nisms other than the global folding and structural changes.

137 Applying the same protein structural analysis to the remaining 18 mutants of FOXA2, we found
138 that this phenomenon occurs especially critical in S231P. Furthermore, the RMSD of the ± 100 aa
139 window was generally higher than that of the ± 50 aa window, suggesting that the mutation has a
140 minor impact on the local amino acid structure near the mutation site, while inducing relatively larger
141 impact on the long-range protein structure.

142 Another possibility is that AlphaFold may ignore local variations while optimizing the global protein
143 structure, thereby leading to structural prediction errors.

144 Using AlphaFold to predict the protein structure reveals the native folding context and provides a
145 structural framework. However, it may fail to present subtle local effects. Therefore, results from
146 AlphaFold prediction were considered carefully to provide insights on the protein structure altering.

147 **3.2 DNA-binding influence analysis**

148 Analysis of the DNA-binding region provided by the PDB revealed that S231 is not located within the
149 DNA-binding region. Thus, it seems unlikely that the functional consequences of the S231P mutant
150 are caused by the direct disrupting of the DNA-binding affinity.

151 We then used PyMOL to examine the mutation site, and found that the S231 residue is surface-
152 exposed, indicating that it is located on the outer surface of the protein. Surface-exposed residues are
153 generally more likely to participate in protein-protein interactions. Therefore, it is possible that the
154 mutation could alter such interactions, and thereby impact the downstream protein functions.

155 On the other hand, although residue S231 is not located within the DNA-binding region, it may alter
156 the local conformation of the protein, indirectly affecting the DNA-binding. This could potentially
157 make the protein-DNA complex more stable and less possible to dissociate, thereby influencing
158 downstream gene regulation.

159 **3.3 Protein-protein interaction analysis**

160 In order to investigate potential mechanisms affecting FOXA2 function, we performed PPI test
161 analyzing protein interactions using STRING. FOXA2 is a key transcription factor driving endoderm
162 differentiation, and it's important in liver and pancreas development. The network result showed that
163 the PPI involved several critical proteins in organ development, including PDX1, HNF1A, ONECUT1,
164 SOX17, and OTX2.

165 These proteins either directly participate in early liver and pancreas development, or controls embry-
166 onic development. For instance, PDX1 and HNF1A are essential for the development of pancreas,
167 ONECUT1 drives liver and pancreatic endocrine differentiation, SOX17 controls endoderm formation
168 and organ lineage specification, and OTX2 directs early embryonic development and brain formation.
169 The role of FOXA2 in this PPI network suggests that disruptions at S231 residue could affect the
170 activity of interacting proteins, and thereby affecting the functioning of those interacting proteins.

171 Furthermore, the S231P mutation may potentially influence differentiation outcomes in iPSC-derived
172 endoderm cells. This PPI result provides mechanistic insight into how S231 mutation in FOXA2 may
173 contribute to disease development, including pancreatic and hepatic diseases observed clinically.

174 4 Methods

175 4.1 AlphaFold resource

176 The protein structure prediction was performed using AlphaFold, obtaining the FOXA2 structure
177 from the AlphaFold Protein Structure Database (UniprotL Q9Y262, model v6). The amino acid
178 sequence of human FOXA2 was used on the AlphaFold to generate the predicted three-dimensional
179 structure, and the model confidence level was estimated based on the given predicted Local Distance
180 Difference Test (pLDDT) score.

181 4.2 Mutant sturcture alignment

182 The FOXA2-DNA complex structure was obtained form the Protein Data Bank (PDB ID: 5X07). We
183 modeled the mutant structure using the AlphaFold server by substituting the serine at residue S231
184 each of the other 19 amino acids, and opened the structure file with PyMOL.

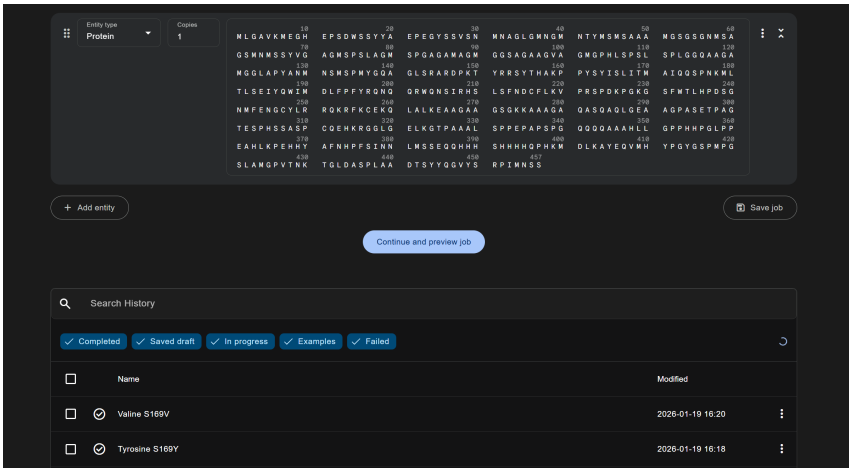


Figure 8: S231 mutant structure prediction with AlphaFold server

185 In PyMOL, we retained either ± 100 aa or ± 50 amino acids surrounding the mutation site, removing
186 all the other residues. Each mutant was then aligned with the wild-type FOXA2, and the RMSD
187 was recorded. The mutation site was showed using sticks presentation to provide a clearer graphic
188 visualization.

189 4.3 DNA-binding site analysis

190 We used PDB to download the crystal structure of the FOXA2 DNA-binding domain (PDB ID:
191 5X07). The structure includes residues 5-98, which is the DNA-binding region of FOXA2. We
192 then used PyMOL to analyze the structure and determine whether the mutation is located within the
193 DNA-binding region or not.



Figure 9: FOXA2 DNA-binding model

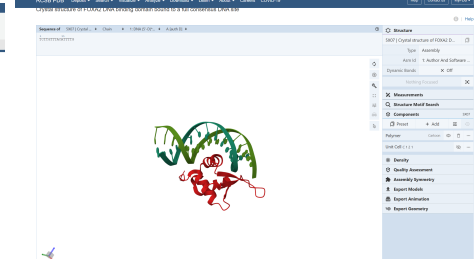


Figure 10: FOXA2 DNA-binding structure

194 4.4 PPI analysis settings

195 PPI analysis was performed using STRING database, version 12.0. The organism was set to Homo
 196 sapiens, with a minimum required interaction score to be 0.400 (medium confidence) and a maximum
 197 number of interactors in the first shell to be 10 interactors. All the active interaction sources were
 198 includes (Textmining, Experiments, Databases, Co-expression, Neighborhood, Gene Fusion, Co-
 199 occurrence). The network edge represent evidence-based functional associations.

Basic Settings

Network type:

☒ full STRING network (the edges indicate both functional and physical protein associations)
 ☐ physical subnetwork (the edges indicate that the proteins are part of a physical complex)

meaning of network edges:

☒ evidence (line color indicates the type of interaction evidence)
 ☐ confidence (line thickness indicates the strength of data support)

active interaction sources:

☒ Textmining ☒ Experiments ☒ Databases ☒ Co-expression
☒ Neighborhood ☒ Gene Fusion ☒ Co-occurrence

minimum required interaction score:

medium confidence (0.400) ±

max number of interactors to show:

1st shell: no more than 10 interactors ±
 2nd shell: - none - ±

Advanced Settings

network display mode:

☐ static png (network is a simple bitmap image; not interactive)
 ☒ interactive svg (network is a scalable vector graphic [SVG]; interactive)

network display options:

☒ Enable node coloring mode
☐ enable 3D bubble design
☐ disable structure previews inside network bubbles
☐ center protein names on nodes
☐ show your query protein names
☐ hide disconnected nodes in the network
☐ hide protein names
☐ 12 protein name font size

UPDATE

Figure 11: PPI analysis settings with STRING Database

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